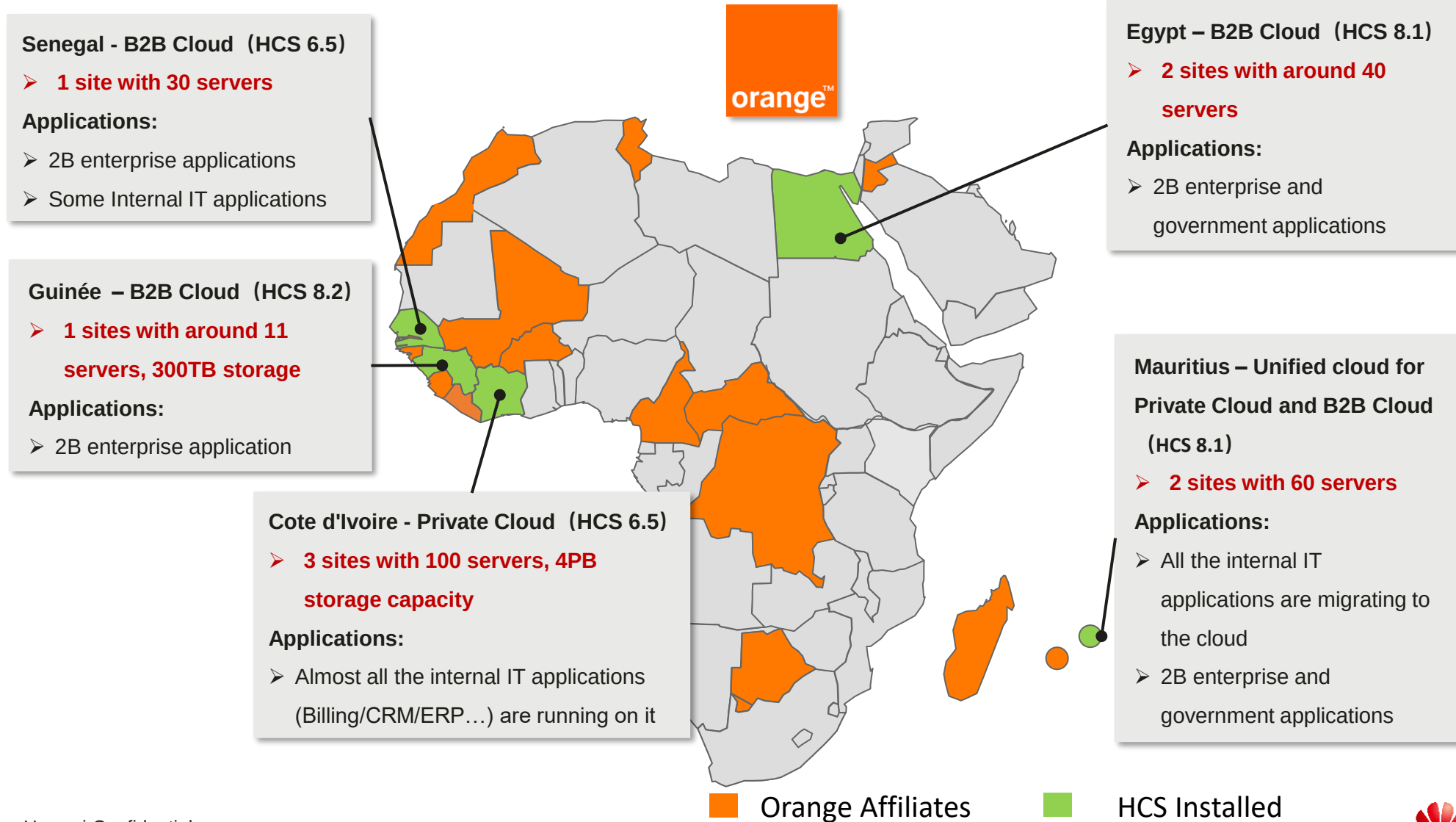
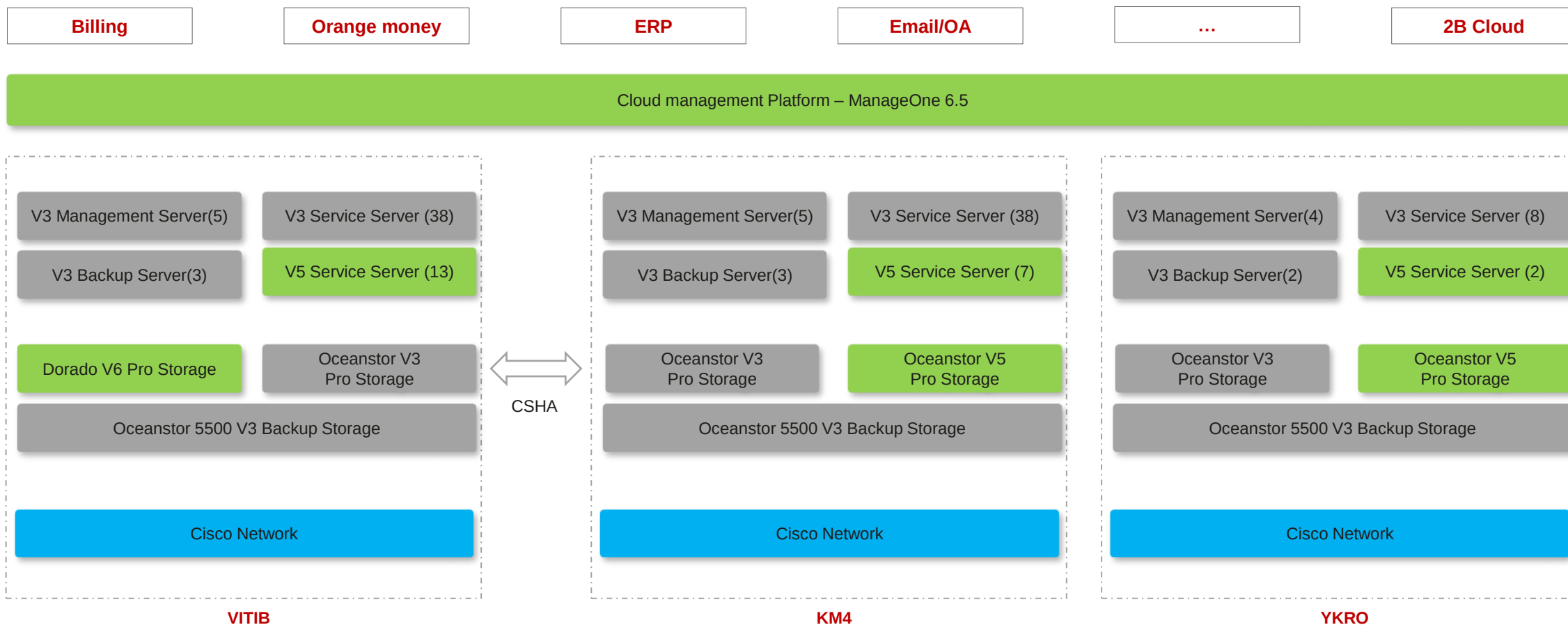


HCS Footprint in OMEA

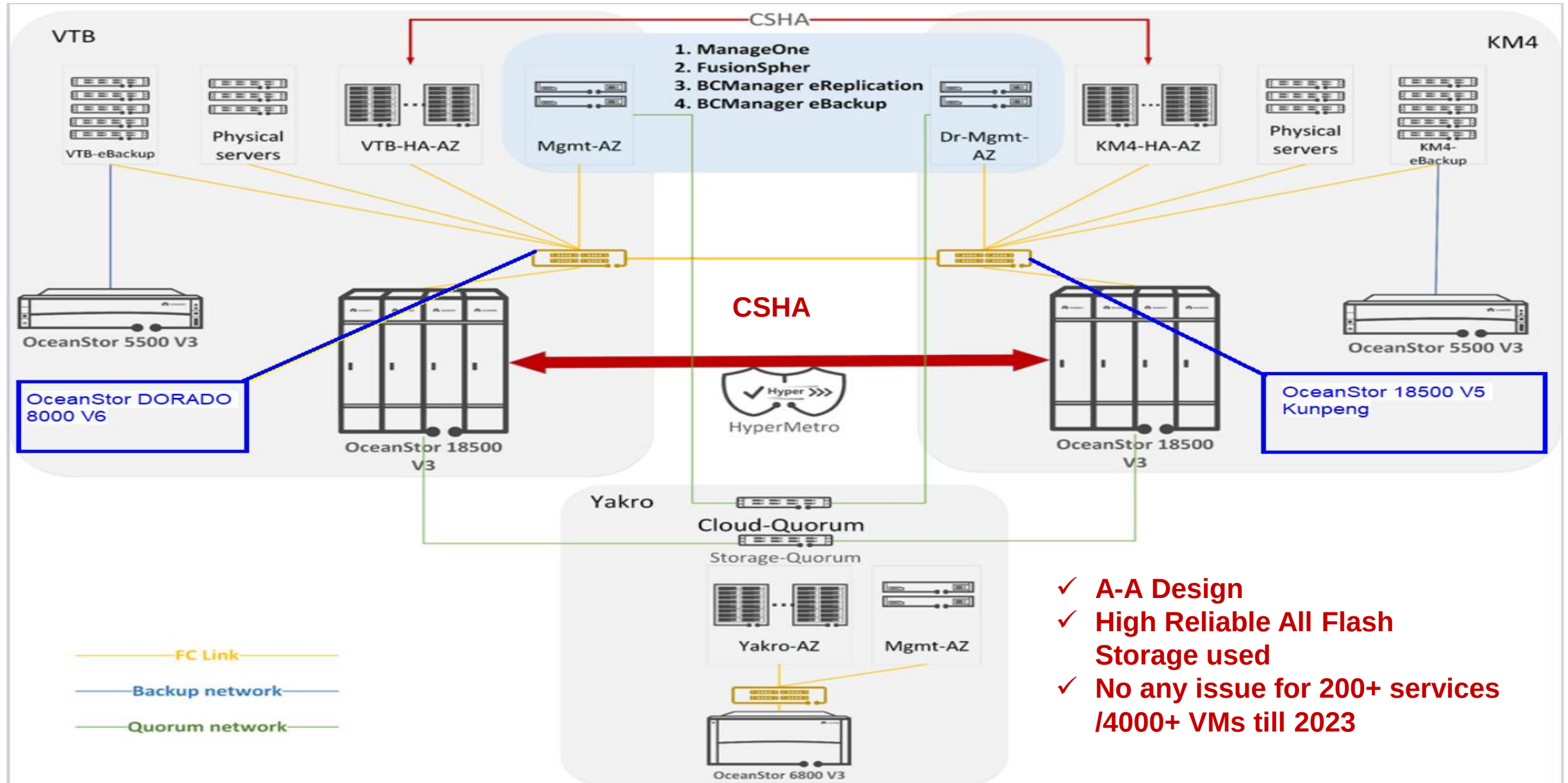


As-Is of OCI Abissa Cloud



- Abissa cloud is **supporting 200+ internal IT App, 10+ critical services (Billing, Orange money, BI, ERP, Email), 10% resource for B2B customer;**
- After deployed in **2017**, Abissa cloud already **run stably almost for 7 years**, without service interruption issue;

OCI Abissa Cloud High Reliable Design



OCI Existing Abissa Platform Summary, Optimize to Support Better Business Development in the Future

Highlights



Most Reliable Cloud

CSHA provides cross-site HA for VMs based on active-active storage, ensuring **zero data loss**



Best Performing Cloud

Huawei **high-end storages** provide the **ultimate performance** for all critical services.



Most Trusted Cloud

Huawei provides **local O&M services** to ensure stable running of Abissa cloud services in the past five years **without service interruption**.

Weakness

× EOS of most hardware

90% hardware will be EOS in 2023, bringing risks to the stability of the cloud platform.

× Limited cloud service

Only 10+ cloud service, without container, data base and other PaaS to meet the future internal service requirement.

× Lack of SDN

SDN is not used and VM networks need to be manually configured, which is error-prone & risks.

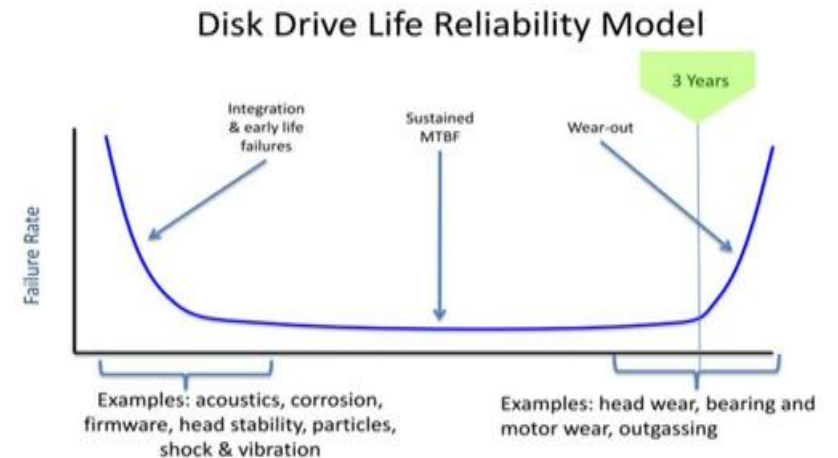
Suggestion

- Replace the EOX hardware to avoid the frequent hardware failure
- Upgrade to the new Abissa with better stability and security for all mission-critical services & 2B cloud service.

OCI Abissa Cloud Needs Upgrade To Ensure Business Continuity

Site	Type	Item	Qty	EOS Time
VTB	ECS Node	CH242 V3	19	2023/12/31
		CH242 V5	5	2025/12/31
		2288H V5	3	2025/12/31
	Production Storage	OceanStor 18500 V3	1	2023/12/31
		Dorado 8000 V6	1	2027/12/31
	Backup System	OceanStor 5500 V3	1	2023/12/31
		RH2288H V3	4	2023/12/31
	Management Node	RH2288H V3	1	2023/12/31
		CH222 V3	4	2023/12/31
KM4	ECS Node	CH242 V3	19	2023/12/31
		CH242 V5	1	2025/12/31
		2288H V5	5	2025/12/31
	Production Storage	OceanStor 18500 V3	1	2023/12/31
		OceanStor 18500 V5	1	2027/12/31
	Backup System	OceanStor 5500 V3	1	2023/12/31
		RH2288H V3	5	2023/12/31
	Management Node	RH2288H V3	1	2023/12/31
		CH222 V3	4	2023/12/31
Yakro	ECS Node	CH242 V3	4	2023/12/31
		CH242 V5	1	2025/12/31
	Production Storage	OceanStor 6800 V3	1	2023/12/31
	Management Node	RH2288H V3	1	2023/12/31
		CH222 V3	3	2023/12/31

```
[Board Properties]
BoardType=STLZ04HD2900
BarCode=2102350FJE10H7000083
Item=02350FJE
Description=OceanStor 18500 V3/18800 V3,STLZ04HD2900,900GB
Manufactured=2017-07-10
VendorName=Huawei
IssueNumber=00
CLEICode=
BOM=
```



- More than **85% servers and storages** are already EOS;
- Both Production/DR **storage usage** rate **more than 90%**, no storage capacity for the **service switchover** when fault occurs;
- The cloud version(**HCS 6.5**) will be EOS in 2025

Abissa Cloud Overall Improvement Suggestion – Smooth Upgrade By 3 Phases

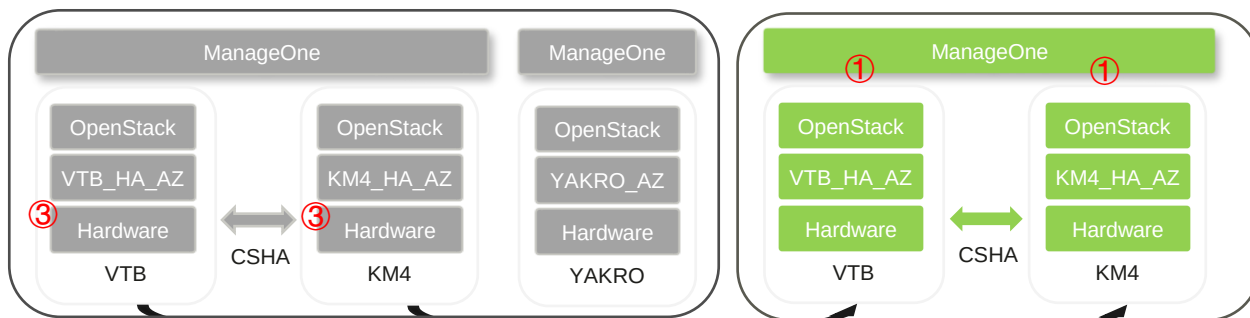
Current

New

Expansion

Current HCS 6.5

New HCS 8.x



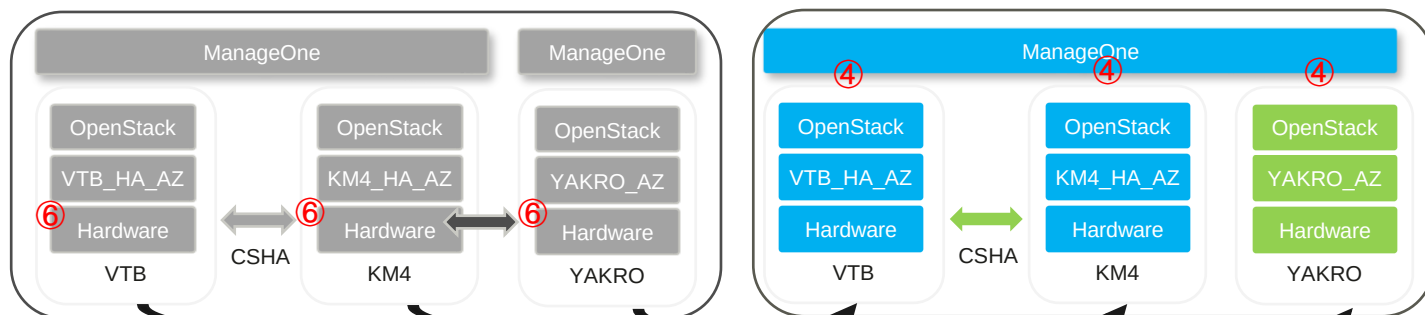
Phase 1: Build HCS 8.2 for KM4 and VITIB

- **Step1:** Build a all new HCS 8.2 with small resources for KM4 & VTB.
- **Step2:** Migrate services from HCS 6.5 v3 servers to HCS 8.2.
- **Step3:** Power off idle v3 servers and storage disks as spare parts.

② Service migration

Current HCS 6.5

HCS 8.x expansion



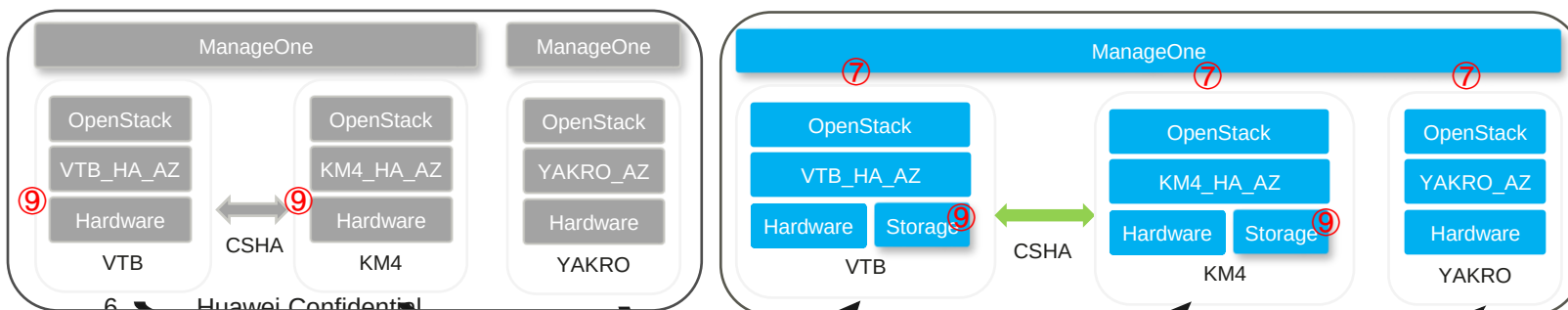
Phase 2: Expand HCS 8.2 for both three site

- **Step4:** Expand HCS 8.2 resources for KM4, VITIB, and build new HCS 8.2 resources for YAKRO.
- **Step5 – Step6:** Repeat step 2 and step 3.

⑤ Service migration

Current HCS 6.5

HCS 8.x expansion



Phase 3: Expand HCS 8.2 & migrate all the service

- **Step7:** Expand HCS 8.2 resource pool for KM4, VITIB and YAKRO sites.
- **Step8 :** Migrate all services from HCS 6.5 to HCS 8.2
- **Step9:** Power off idle v5 and v6 storage of HCS6.5 for HCS 8.2 expansion.

⑧ Service migration

Abissa Upgrade Hardware Plan Overview By Phases

Site	Phase 1(2024)	Phase 2(2025)	Phase 3(2026)	Remark
VTB	5*2288H			Cloud management server
	3*2288H			SDN server
	2*2288H			Backup server
	4*2288H	7*2288H	7*2288H	App server
	2*2288H			Arbitration server
	1* Dorado 8000 with 90TB	Expansion 150TB	Expansion 150TB	Production storage
	1* Oceanstor 5610 v6 with 100TB	Expansion 160TB	Expansion 160TB	Back storage
	network switch & firewall		Switch expansion	Network switch and firewall
	FusionSphere & Mangeone	Expansion FusionSphere & Mangeone	Expansion FusionSphere & Mangeone	Huawei cloud software
KM4	5*2288H			Cloud management server
	3*2288H			SDN server
	2*2288H			Backup server
	4*2288H	6*2288H	6*2288H	App server
	1* Dorado 8000 with 90TB	Expansion 130TB	Expansion 130TB	Production storage
	1* Oceanstor 5610 v6 with 100TB	Expansion 140TB	Expansion 140TB	Back storage
	network switch & firewall		switch expansion	Network switch and firewall
	FusionSphere & Mangeone	Expansion FusionSphere & Mangeone	Expansion FusionSphere & Mangeone	Huawei cloud software
Yakro		3*2288H		Cloud management server
		2*2288H		SDN server
		3*2288H	4*2288H	App server
		1* Dorado 6000 with 25TB	expansion 30TB	Production storage
		network switch & firewall		Network switch and firewall
		Expansion FusionSphere & Mangeone	Expansion FusionSphere & Mangeone	Huawei cloud software

The comparison between TCO of New Abissa Upgrading offer and Existing Abissa Platform

Pervious PO Amount of Existing Abissa Platform :

Year	PO (EUR)
2017	3,400,000.00
2019	135,973.33
2020	332,389.93
2021	413,888.88
2022	338,377.78
2023	622,333.74
SUM	5,242,963.66

Offer Amount of New Abissa Platform Upgrading:

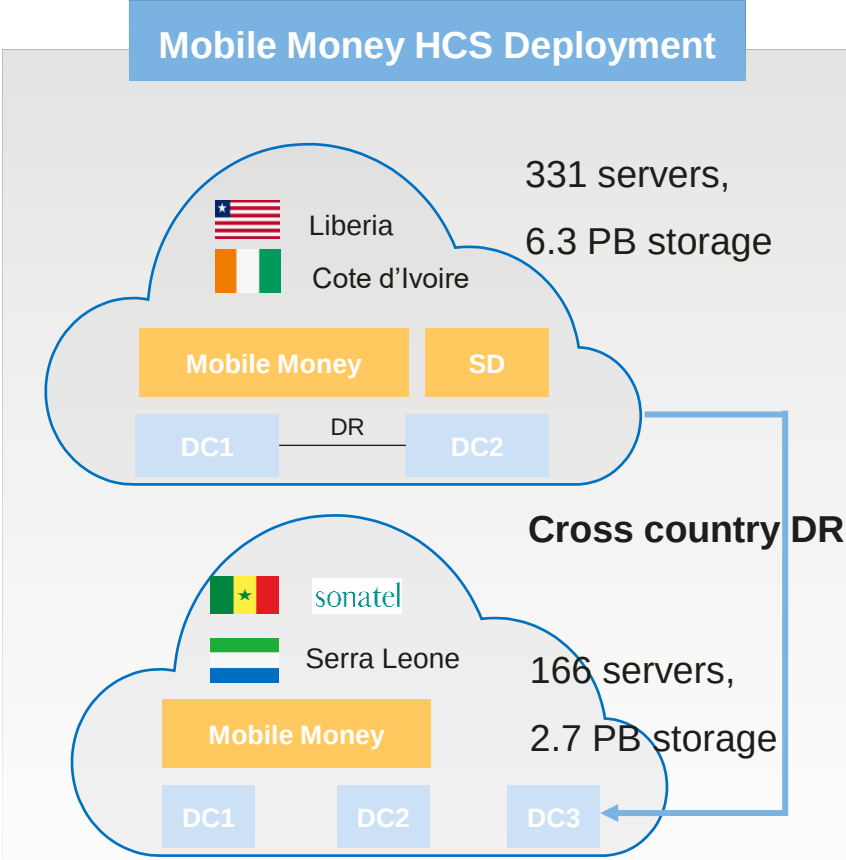
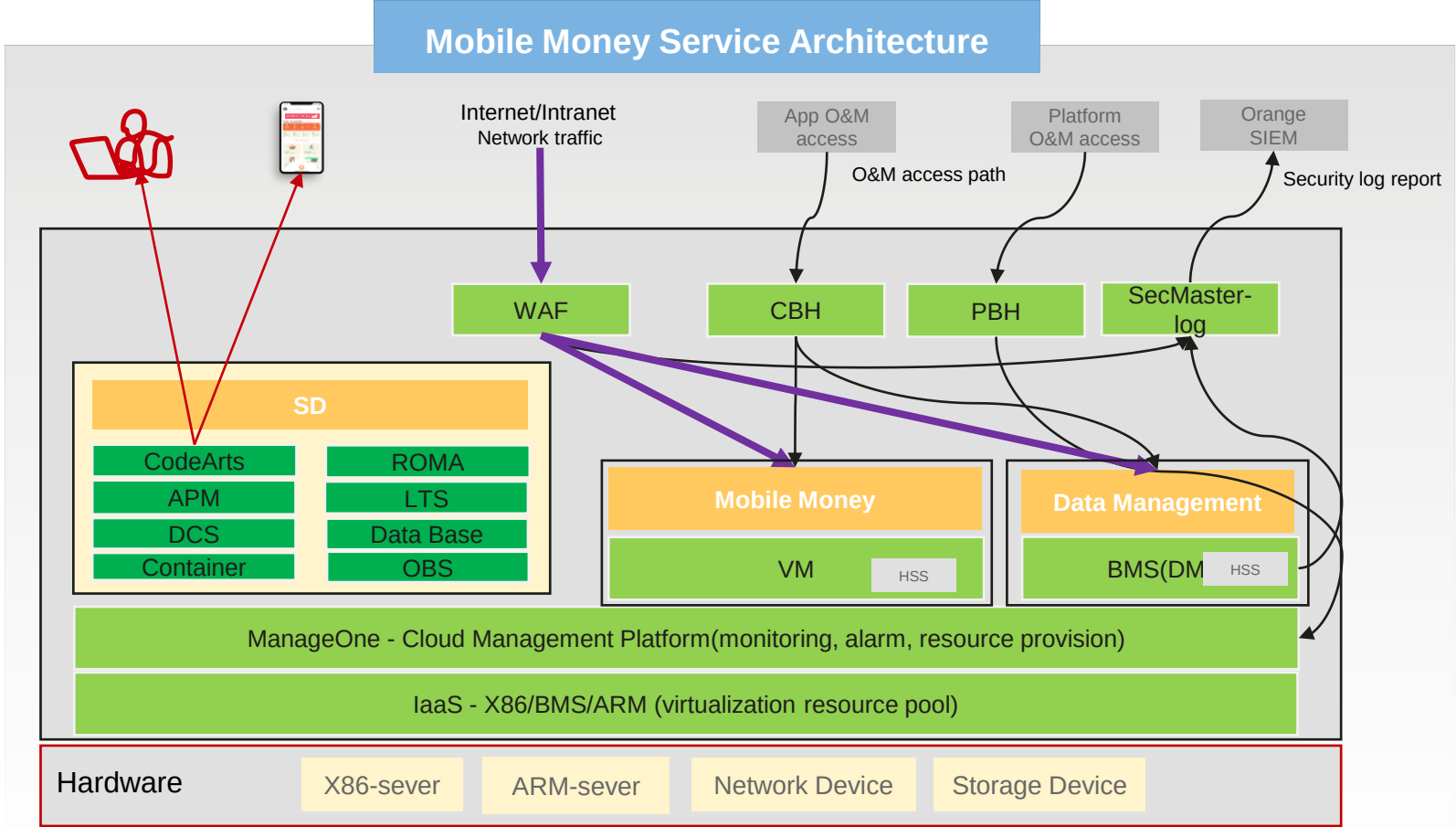
Year	PO (EUR)
2024	901,428.57
2025	1,159,428.57
2026	1,042,571.43
2027	544,000.00
2028	624,571.43
SUM	4,272,000.00

The amount of New Abissa Platform Upgrading Offer is less than previous TCO investment of Existing Abissa Platform.

Next Step:

- Align on the technical and commercial offer and move on for the upgrade

Mobile Money: HCS Cloud Service License function description



Highlights of HCS for MM

High Security: comprehensive security services

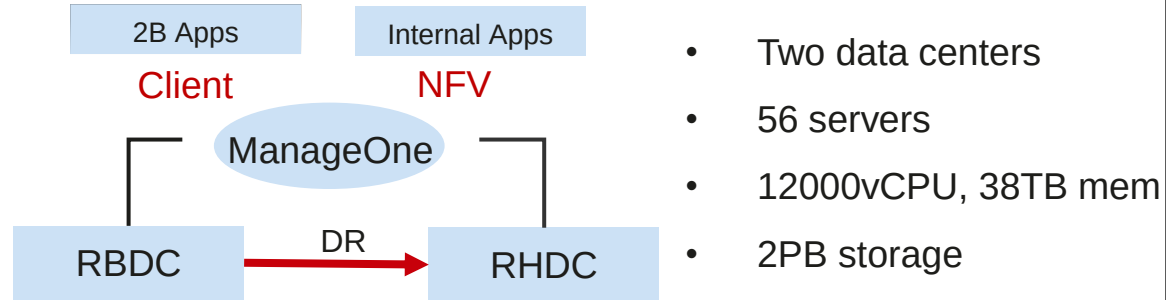
More Agile: Fast TTM for new apps with PaaS

Easy O&M: unified management for all infrastructure

Low Cost: one platform shared by different apps

Mauritius HCS platform Overview and Evolution Suggestions

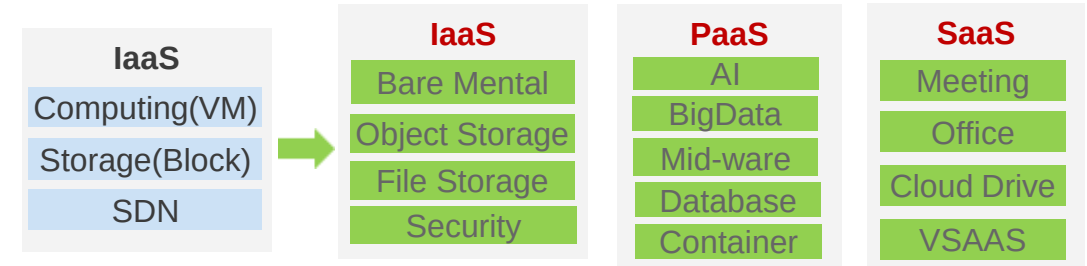
• HCS: Unified cloud for internal and 2B business



Analysis: Compute and Storage reaching 70% threshold

RBDC	Compute	NFV GENERAL & DEDICATED	94%
	Storage	NFV FSB SSD & DORADO NVME	94%
RHDC	Compute	NFV DEDICATED	70%
		CLIENT GENERAL & DEDICATED	87%
	Storage	Backup Stor OS5500	100%
HA Reservation		Disabled	➔ 10% Reservation

Evolution: Expansion needed and New Cloud Services Roll Out for NFV and Client



Internal IT:

All internal apps migrate to HCS NFV, 100 VMware VMs migration is done

B2B Business:

Mauritius Telecom cloud GTM is done, 200+ government and enterprise customer joined, 15+ customers are waiting for the resource expansion



Internet

11 clients, 400K USD/M



Smart Port

2 clients, 400K USD/M



Healthcare

2 clients, 200K USD/M



Blue Economy

1 clients, 400K USD/M

2B HCS Deployment and Evolution in Egypt and Guinea

Egypt HCS Multi-phase Evolution

100+ 2B users


Government


Chinese Enterprise

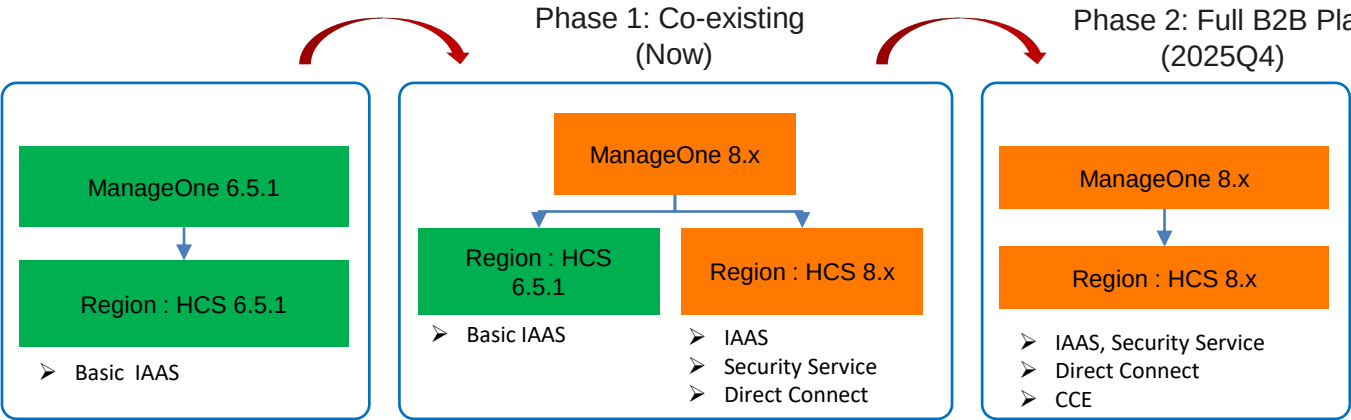

Other Enterprise

HCS 6.5.1

- Constrained resource pool limiting new service deployment (<200 VM)
- Contained only in basic IaaS
- Outdated, less secure

HCS 8.x

- Enhanced B2B, Agile on-boarding of new service (~500VM)
- New security and private line services cater for enterprise sales
- Tech Refresh, level-up security posture



Guinea 2B HCS Evolution

Smart Mining
(100M+, 80% Chinese enterprises)



Smart Education
(80M+, Cloud/Network/App)

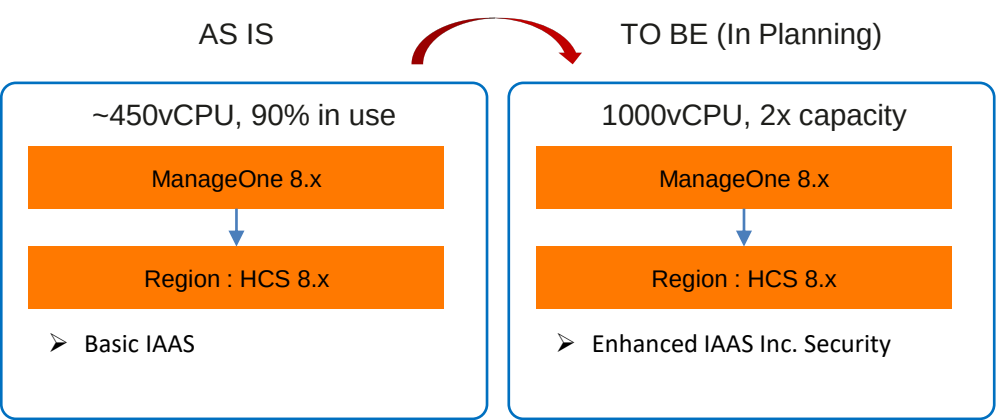


2B Enterprise

- Cloud + Production Monitoring App
- Cloud + Campus Network

Plus 2B SME

- SME Digital Package (Cloud Meeting, Cloud PoS etc.)
- Cloud Drive for SME/SOHO



Key Affiliates are complain with VMware Price Increasing



VMware Changes:

1. Independent sales to **bundle sales**, license+SnS model to **annual subscription model**
2. **Price increasing** due to unit price increasing and bundle sales, customers are forced to use the software package

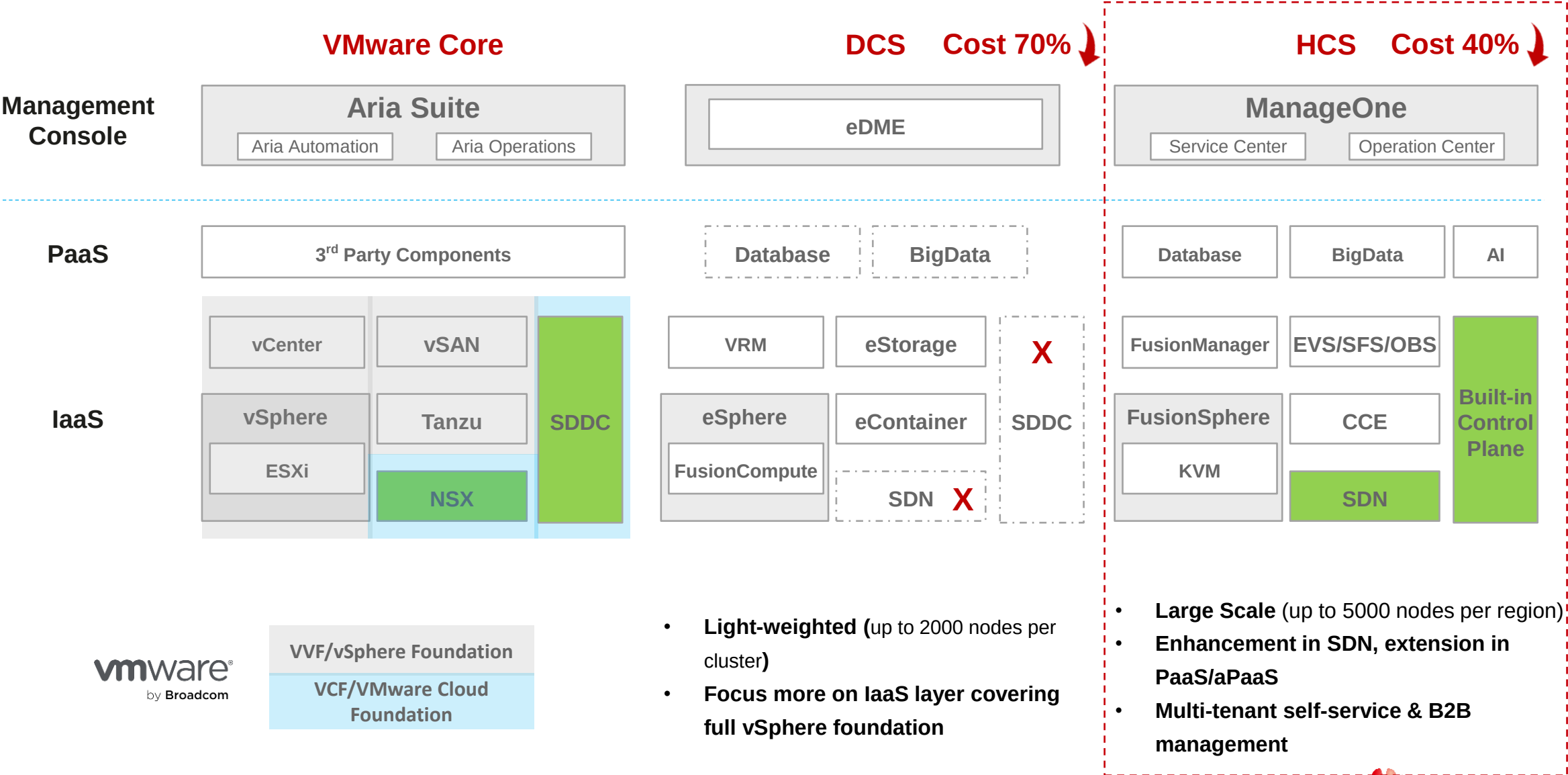
Impacts on Orange: 4-5X Price Increasing, existing contract only extend to 2025

OBS Already take legal action with VMware:

<https://alloforfait.fr/business/news/135557-orange-attaque-vmware-justice-suite-fortes-augmentations-prix.html>

Affiliates	 Egypt	 Senegal	 Congo	 Guinea
Existing Environment	VMware virtualization 200+ servers, 2500+ VMs	VMware Virtualization + 2B Cloud 80+ servers, 1000+ VMs	VMware virtualization 60+ servers, 1000+ VMs	VMware virtualization 30+ servers, 500+ VMs
Pain Points	Frame contract expire in 2025 Price Increase 4X	Frame contract expire in 2025 Price Increase 5X	Frame contract expire in 2025 Price Increase 5X , need 2B cloud	Frame contract expire in 2025 VMware version is EOL
Plan	POC Ongoing	RFP Ongoing	POC with Huawei in 2025	Project planned in 2025

VMware alternative solution: Huawei HCS for cloud, DCS for virtualization



HCS is taking lead in IaaS & PaaS and improving the Ecosystem

IaaS & PaaS

(Huawei is equal with VMware in IaaS and far better in advanced services)

<u>Key Characters</u>		<u>VMware</u>	<u>HCS</u>
1.	Basic IaaS	10+	10+
2.	Bare Metal Server	Weak	✓
3.	Object Storage	X	✓
4.	Rapid Elasticity	✓	✓
5.	Enhanced Virtual Network / SDN	✓	✓
6.	Backup of own intellectual property	X	✓
7.	Container & Microservice	✓	✓
8.	Database & Bigdata	Weak	Strong
9.	Application Integration Framework	Weak	Strong
10.	AI training and inference	Weak	Strong

Management & Ecosystem

(Huawei is better in B2B and slightly weaker in datacenter automation)

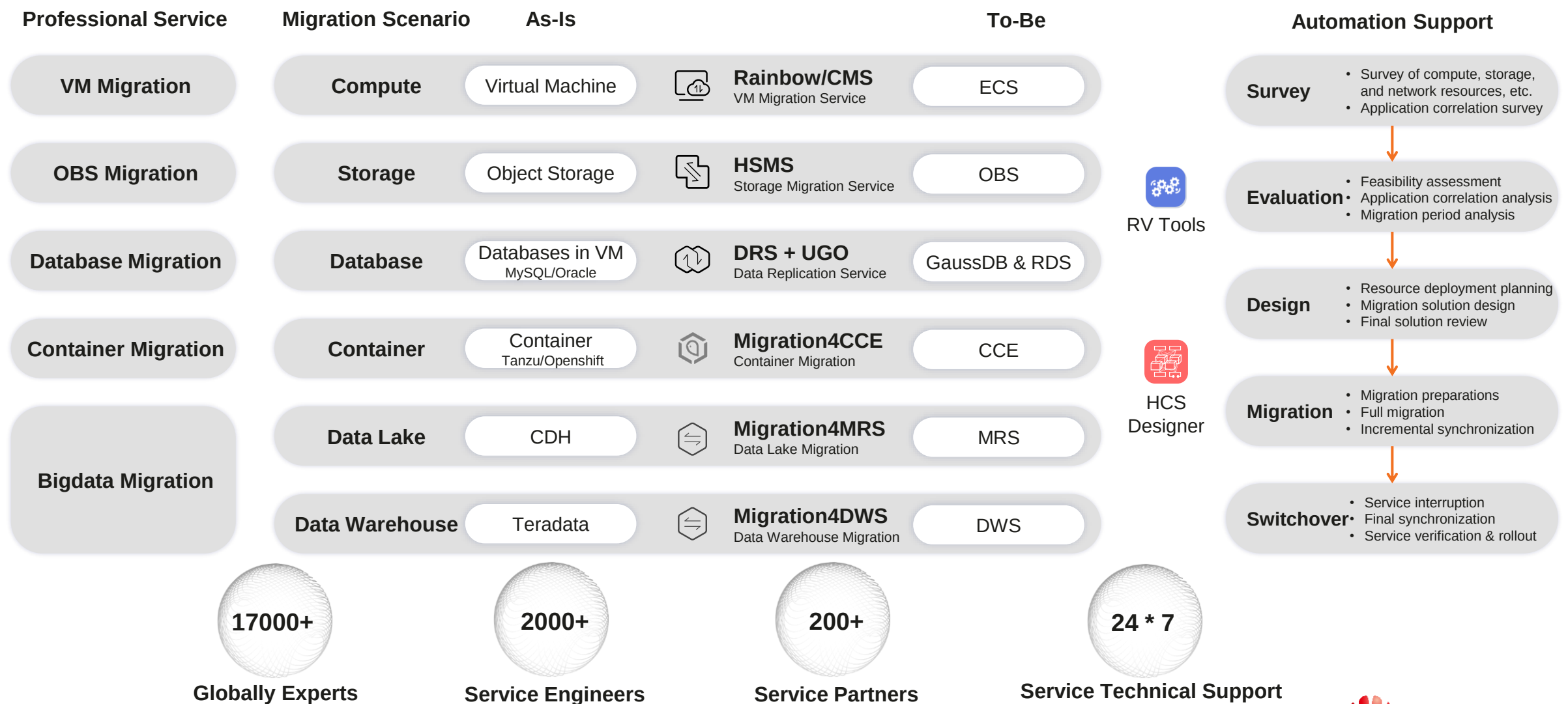
<u>Key Characters</u>		<u>VMware</u>	<u>HCS</u>
1.	Basic O&M	✓	✓
2.	Multi-tenant / VDC	✓	✓
3.	Unified Catalog	✓	✓
4.	B2B Quota, Billing, Process Flow	X	✓
5.	Enhanced Automation	✓	Moderate
6.	FinOps	✓	X
7.	Cloud Federation	Moderate	Strong
8.	Partner Ecosystem	✓	Moderate

Partner On-boarding

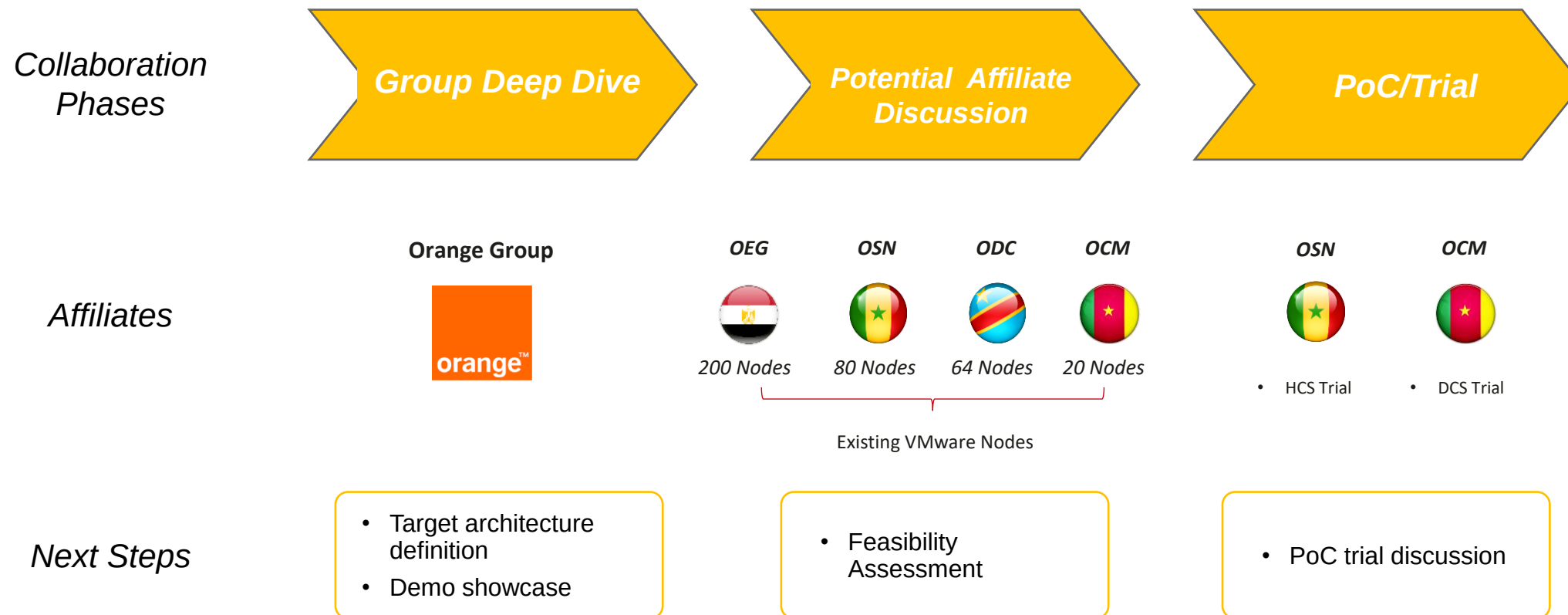
Consultancy Partner

Technology Partner

8 Migration Tools for 6 Migration Scenarios



Huawei Suggestions for Collaboration with Orange



Cooperation Summary

1. OCI Abissa HCS upgrade for business continuity
2. VMware: **OGN** trial with existing HCS, **OCM** trial with Huawei DCS
3. Big Data: **OSN** trial with existing HCS